

GENE TECHNOLOGY

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- **Evaluation**

- Experiment: 30%
- Middle term exam (written test): 10%
- Seminar: 10%
- Final exam (written test): 50%

- **References**

- Lecture slides
- [1] Old, R.W. & Primrose, S.B. Principles of Gene Manipulation. *Blackwell Scientific Publication*, 2008.
- [2] Sambrook, J., Fritsch, E.F. & Maniatis, T. (2001) Molecular Cloning, A Laboratory Manual, 3rd ed., Volume 1, 2 and 3. *New York: Cold Spring Harbor Laboratory Press*.
- [3] Newton, C.R. & Graham, A.PCR. *Oxford: BIOS Scientific Publishers Limited*, 1994.
- [4] Bernard R., Glick, Jack J. Pasternak. Molecular Biotechnology – Principles and Application of Recombinant DNA. *American Society of Microbiology, Washington, DC*, 2010.
- [5] Watson JD, Baker TA, Bell SP, et al. (2014)– Molecular Biology of the Gene. Pearson Education, Inc., 7th ed., Cold Spring Harbor Laboratory Press.
- [6] Gerstein AS- Molecular Biology Problem Solver (A laboratory Guide). Wiley-Liss, Inc., 2001.
- [7] Gellissen G (2005) Production of recombinant proteins. Willey-VCH Press.
- Others

- **Chapter 1: Nucleic acids extraction and purification**
- **Chapter 2: PCR and electrophoresis**
- **Chapter 3: Vectors and cloning**
- **Chapter 4: Blotting methods**
- **Chapter 5: Sequencing**
- **Chapter 6: Gene transfer in microorganisms**
- **Chapter 7: Gene transfer in other organisms**
- **Chapter 8: Recombinant proteins and detection methods**
- **Seminar: Applications of gene technology**

Chapter 1

Nucleic acids extraction and purification

I. Introduction

II. Extraction of nucleic acids

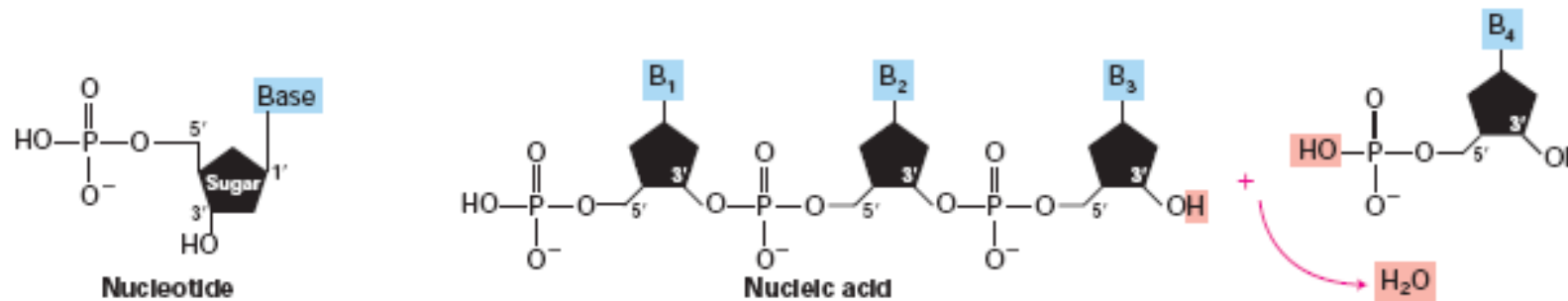
III. Quantification and purification of nucleic acids

IV. Plasmid extraction (experiment)

I. Introduction

Nucleic acids

- Genomes: prokaryote, eukaryote, virus
 - prokaryote, eukaryote: DNA
 - virus: DNA or RNA
- Monomers of nucleic acids



I. Introduction

Reasons to need nucleic acids extracted?

- **Detection**

....virus, bacteria, fungi,...

- **Diagnosis of disease**

mutation in genes → disease

- **In general testing**

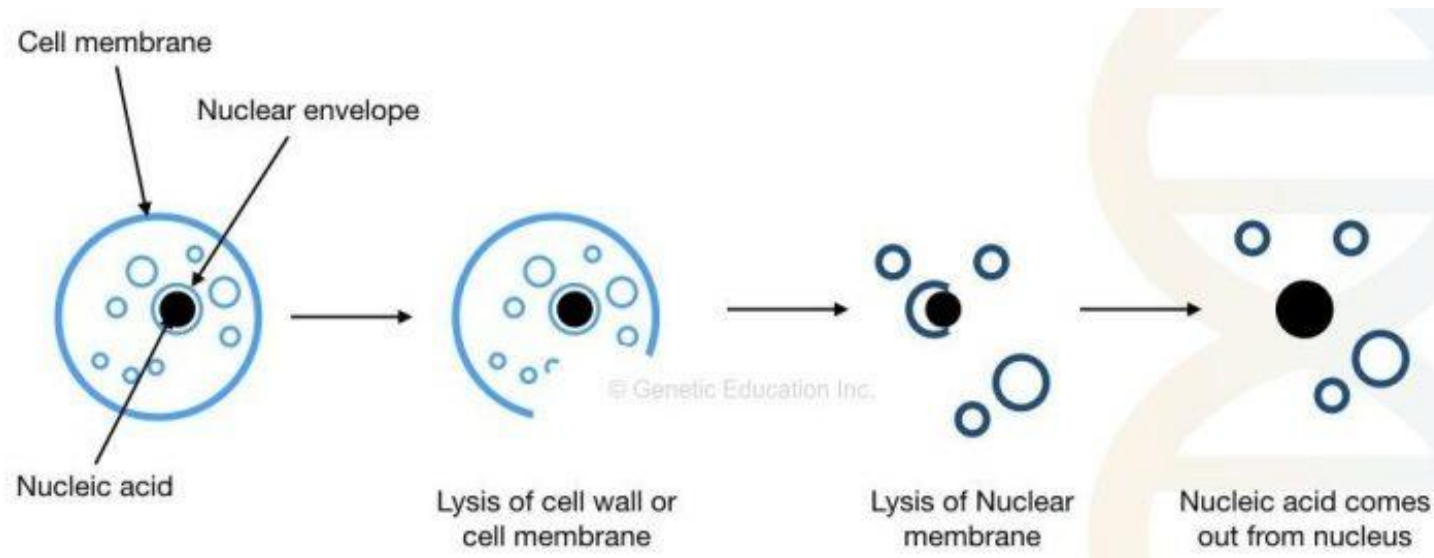
- **In ensuring food safety and environment protection**

- **In R & D**

I. Introduction

Extraction of nucleic acids

- Extraction of nucleic acids is a fundamental laboratory process used to isolate DNA or RNA from biological samples such as blood, tissues, cells, or microorganisms.
- Cell wall: animal cell v.s. plant cell, bacterial cell
- Cell membrane: prokaryote v.s. eukaryote
- Cell components: nucleic acids, proteins, lipids, ions,....



I. Introduction

Extraction of nucleic acids

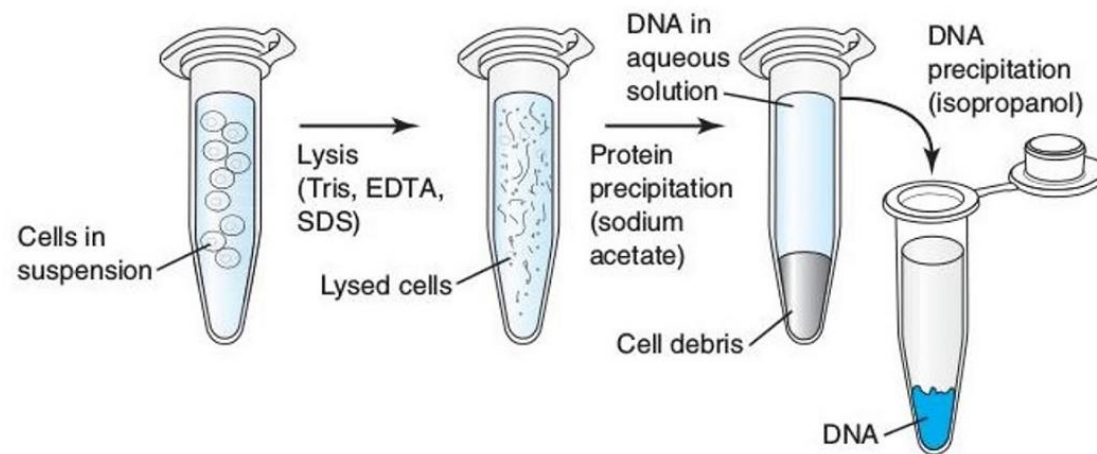
- Nucleic acids: DNA (or RNA), plasmids
 - Nucleic acids have relatively similar biochemical properties
- The extraction methods applied to DNA or RNA are similar...

I. Introduction

Extraction of nucleic acids

- **Main steps**

1. Sample collection and preparation
2. Cell lysis
3. Separation of nucleic acids with other cell components
4. Purification of nucleic acids



I. Introduction

Classification of methods

- **Liquid-phase DNA extraction method**
 - Phenol-chloroform method
 - Enzyme (Proteinase K) method
- **Solid-phase DNA extraction method**
- **Magnetic Beads**

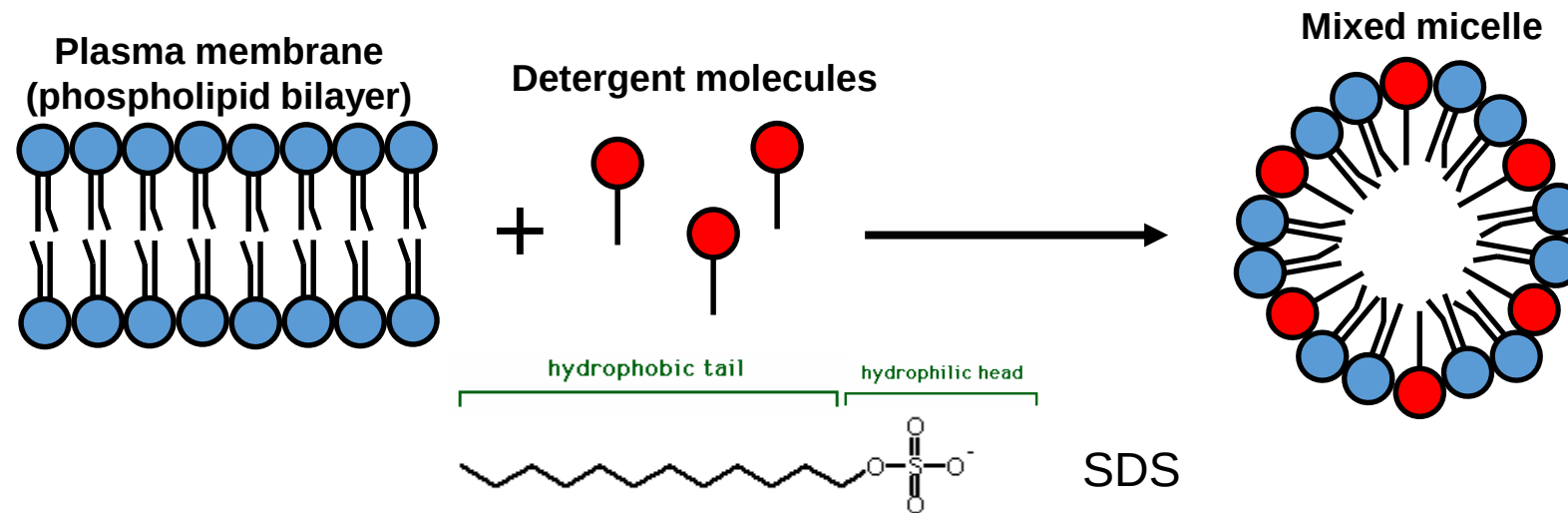
Convenient and automatable for high-throughput extraction

II. Extraction methods

II.1. Phenol-chloroform method

Step 1: Cell lysis

SDS: Sodium dodecyl sulphate (detergent)



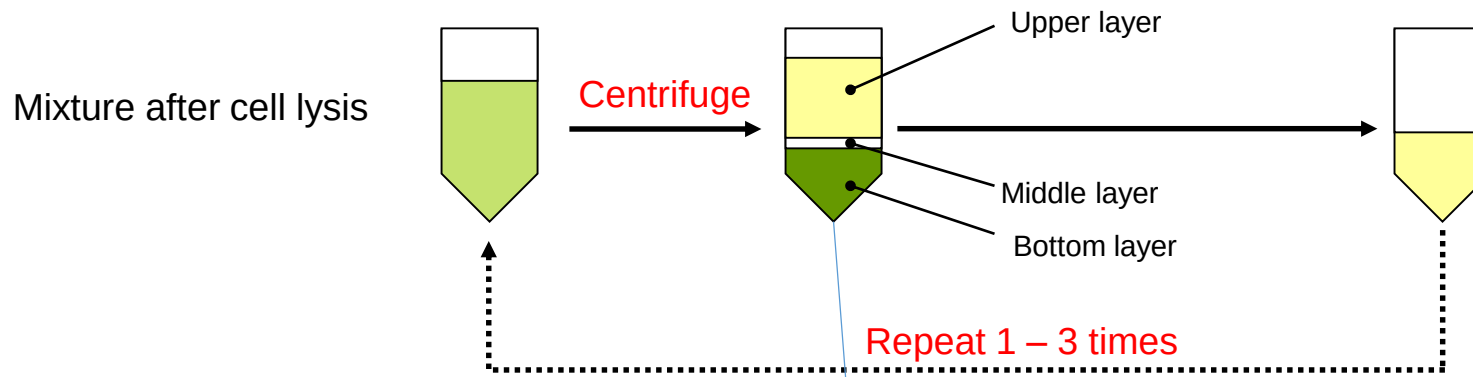
Phenol-chloroform: denature proteins

Tris: stabilize pH

EDTA (Ethylenediaminetetraacetic acid): inhibit Dnase/RNase

II.1. Phenol-chloroform method

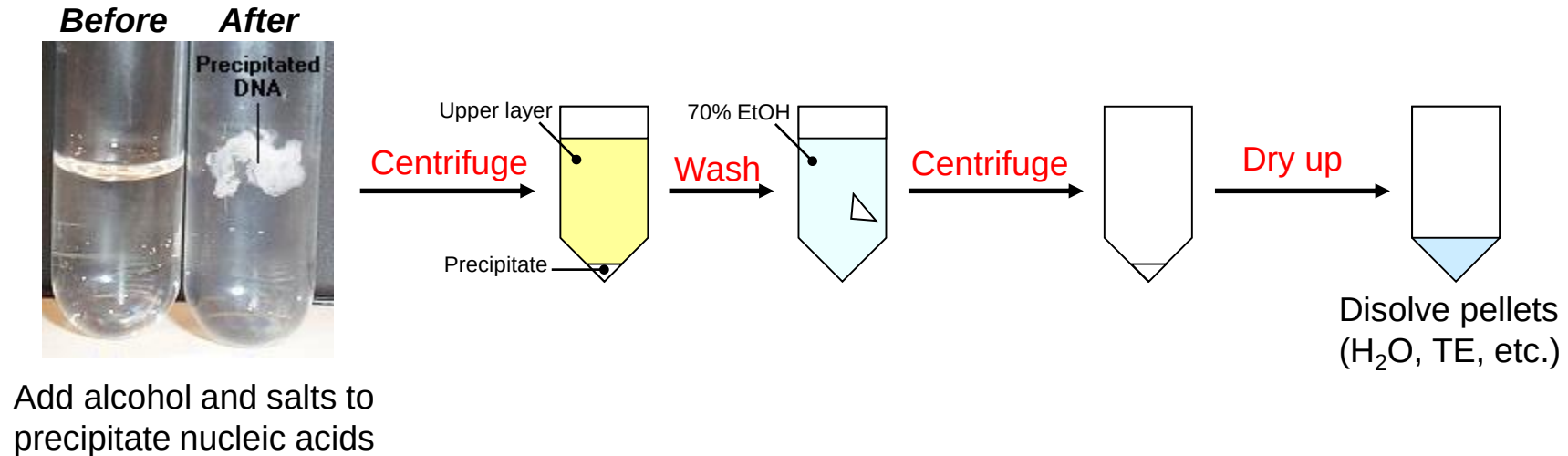
Step 2: Separation of nucleic acids



Upper layer: Dissolved molecules (including nucleic acids)
Middle layer: Denatured proteins and others
Bottom layer: Organic solvents

II.1. Phenol-chloroform method

Step 3: Precipitation of nucleic acids



Nucleic acids stored in TE buffer (Tris-EDTA Buffer); -30 °C

II.2. Enzyme method

Lysis buffers includes:

- Tris
- EDTA
- SDS
- Enzyme proteinase K (instead of phenol-chloroform)

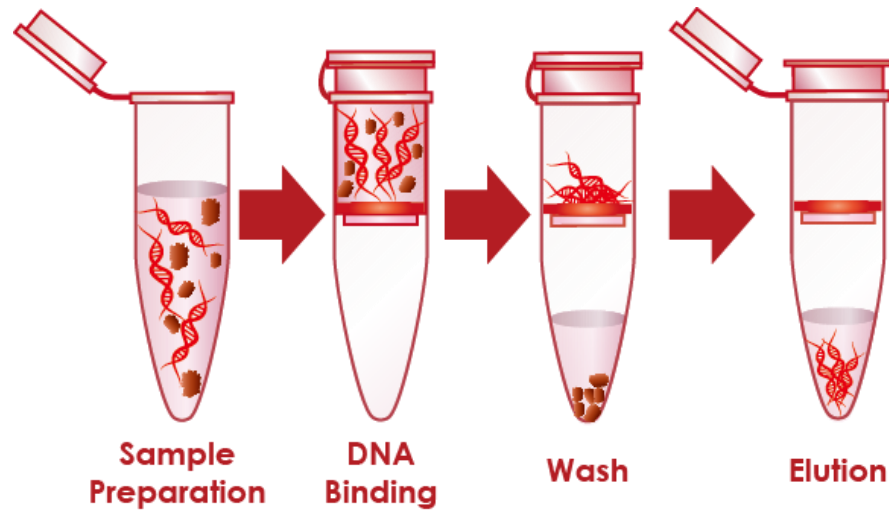
Proteinase K denatures and degrades proteins in samples

II.3. Solid-phase DNA extraction method (using column)

- **Silica column**

- **Steps:**

- Cell Lysis
- Binding
- Washing
- Elution



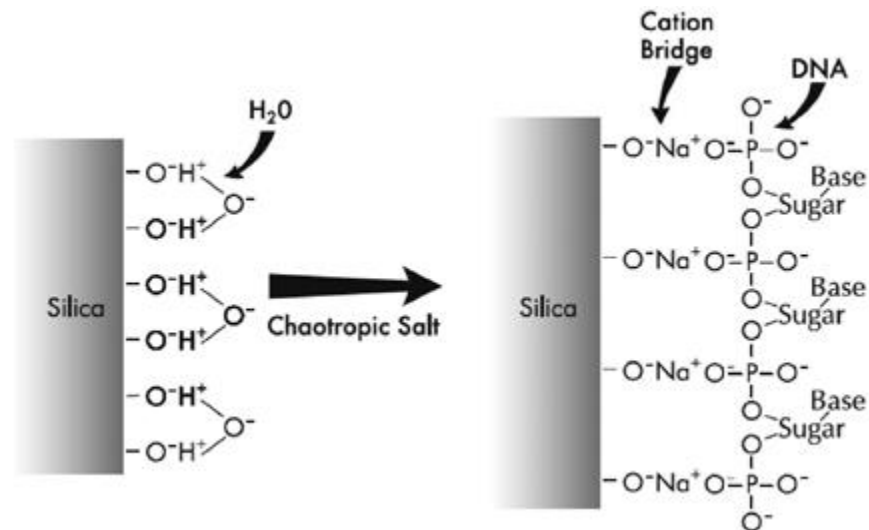
II.3. Solid-phase DNA extraction method (using column)

- **Cell lysis**

- The sample (e.g., cells, tissue) is lysed to release the DNA
- Proteinase K

- **Binding**

- The lysate is mixed with a buffer containing chaotropic salts, which disrupt hydrogen bonds → This allows the DNA to bind to the silica matrix in the spin column



II.3. Solid-phase DNA extraction method (using column)

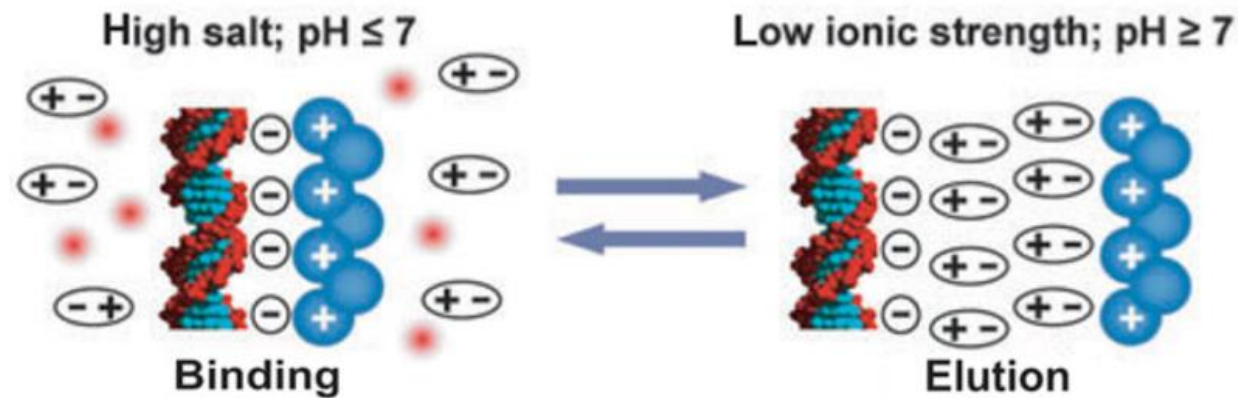
- **Washing**

Wash buffers, often containing alcohol, are used to remove contaminants and unbound material (protein, polysaccharides,...) from the column.

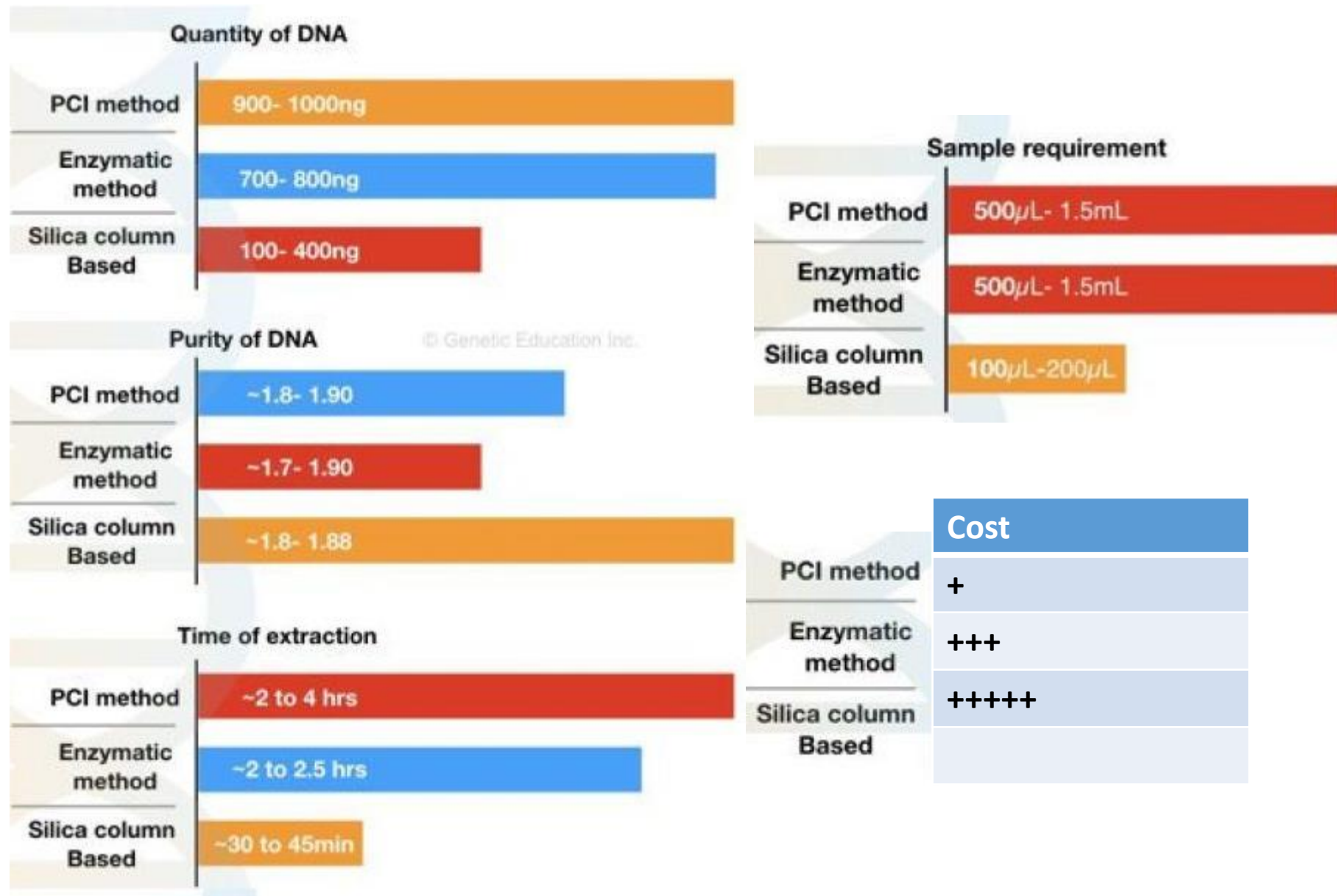
II.3. Solid-phase DNA extraction method (using column)

- **Elution**

A low-salt buffer (often containing Tris-EDTA) is added to the column to disrupt the DNA-silica interaction, releasing the purified DNA into a collection tube.

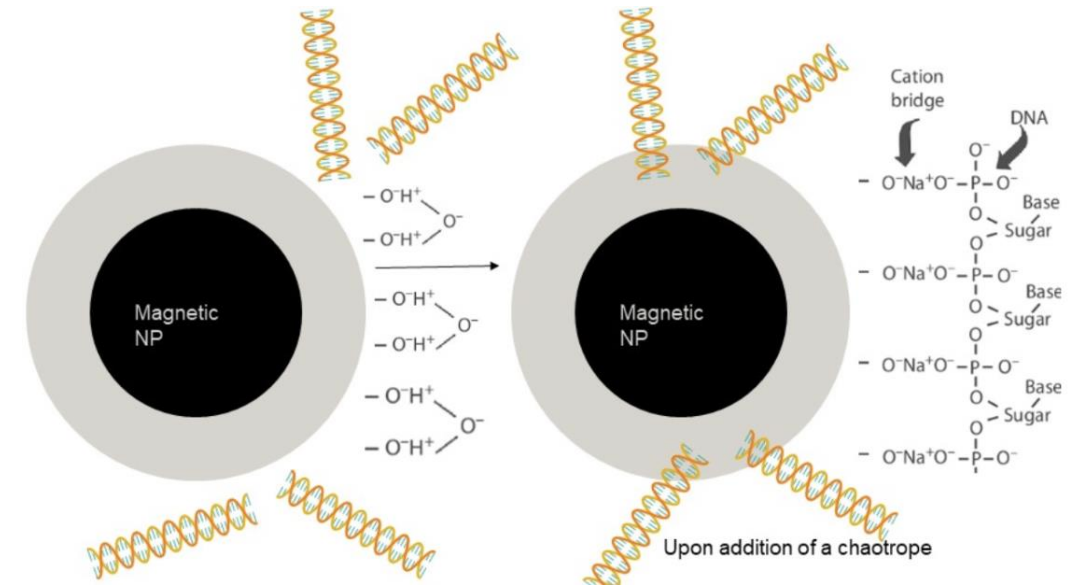


Comparison of the methods



Magnetic Beads–Based Nucleic Acid Extraction

- **Magnetic beads** are tiny particles (often coated with silica or other functional groups) that can selectively bind nucleic acids under specific buffer conditions.
- When a lysis buffer containing chaotropic salts is added, nucleic acids are released from cells and adsorb to the bead surface.
- A magnet is then applied to immobilize the beads, allowing contaminants such as proteins, lipids, and salts to be washed away easily.
- Finally, purified nucleic acids are eluted in a low-salt buffer or water.



Magnetic Beads–Based Nucleic Acid Extraction

This method offers several advantages:

- **High efficiency and purity:** Produces nucleic acids suitable for sensitive downstream applications like PCR, qPCR, sequencing, and cloning.
- **Safety:** Avoids hazardous chemicals such as phenol and chloroform.
- **Scalability and automation:** Compatible with robotic platforms for high-throughput laboratories.
- **Versatility:** Applicable to a wide range of samples, including blood, tissues, swabs, and microbial cultures.

→ Magnetic beads–based extraction has become a preferred method in clinical diagnostics, molecular biology, and forensic science, particularly where speed, reproducibility, and automation are critical.

III. Nucleic acids quantification

Nucleic acid quantification is the process of measuring the amount and purity of DNA or RNA in a sample.

Why it is important?

- **Downstream Applications:** Many molecular biology techniques, like PCR, sequencing, and cloning, rely on precise nucleic acid concentrations.
- **Reproducibility:** Accurate quantification ensures that experiments are reproducible and that results are reliable.
- **Avoiding Errors:** Inaccurate quantification can lead to failed experiments, variability in results, and wasted resources.

III. Nucleic acids quantification

The Beer-Lambert's equation

$$A = \epsilon cl$$

C= nucleic acid concentration ($\mu\text{g/mL}$)

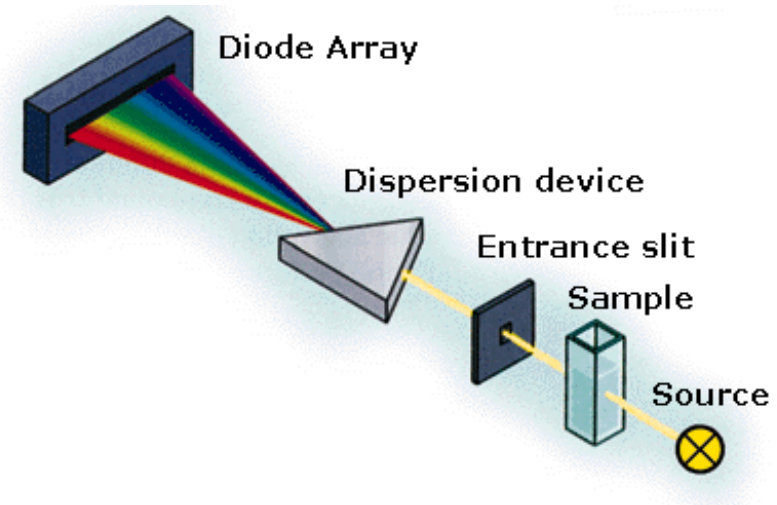
A= UV absorbance in absorbance units (-)

ϵ = extinction coefficient (độ hấp thụ quang riêng)
($\text{mL}/\mu\text{g}\cdot\text{cm}$)

$\epsilon = 0.02$ for dsDNA

$\epsilon = 0.025$ for RNA

l= light path in cm (cm)



$$[\text{dsDNA}] \approx A_{260} \times (50 \mu\text{g/mL})$$

$$[\text{RNA}] \approx A_{260} \times (40 \mu\text{g/mL})$$

Purity of nucleic acids

Nucleic acids (DNA and RNA) strongly absorb ultraviolet (UV) light at a wavelength of 260 nm; Proteins: absorb strongly around 280 nm; Organic solvents and certain salts absorb at around 230 nm.

A pure DNA sample: A_{260}/A_{280} ratio of ~ 1.8 and A_{260}/A_{230} ratio $\sim 1.8 - 2.0$

A pure RNA usually has an A_{260}/A_{280} ratio of ~ 2.0

IV. Plasmid extraction (experiment)